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International Mathematics Assessments for Schools

2021 ~ 2022 Middle Primary Division 2nd Round Paper

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

2021 ~ 2022 Middle Primary Division 2nd Round Paper

Questions 1 to 5, 4 marks each

1. Determine the logic of the patterns on Figure A and use that knowledge to find which unit square should Figure B be placed into.

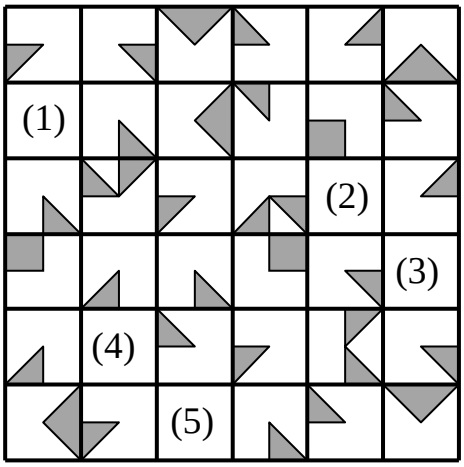


Figure A



Figure B

（A）1　　（B）2　　（C）3　　（D）4　　（E）5

Answer：_____

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2. Three farmers Alvin, Benny and Charlie went out to harvest chestnuts where Alvin was able to get 116 chestnuts, Benny 112 chestnuts and Charlie 96 chestnuts. On their way home, one of them gave $\frac{1}{4}$ of his chestnuts to one of the other two. Then, another one gave $\frac{1}{4}$ of his chestnuts to one of other two. Finally, the third farmer gave $\frac{1}{4}$ of his chestnuts to another farmer. After all of these, the total number of chestnuts of all three farmers are the same. At the start, which farmer gave out $\frac{1}{4}$ of his chestnuts to which farmer?

（A）Alvin to Benny　　（B）Alvin to Charlie　　（C）Benny to Alvin
（D）Benny to Charlie　　（E）Charlie to Alvin

Answer：_____

3. Peter will be attending a two-day summer camp, where he plans to wear different colour of socks in both days. He keeps all of his socks in a messy non-transparent drawer under his bed and it is known that he has 8 black socks and 6 white socks. If he takes out 1 sock at a time, how many socks does he need to get to be sure that he gets 2 pairs of socks of different colour?

（A）4　　（B）6　　（C）8　　（D）10　　（E）12

Answer：_____

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4. In the sequence,

1, 2, 4, 8, 6, …

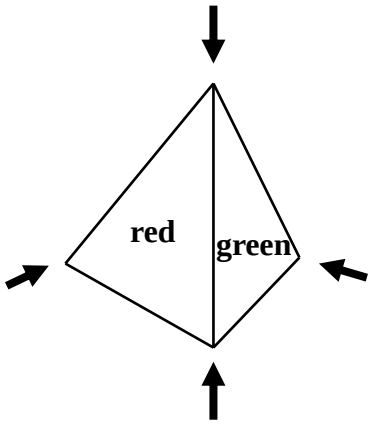
Starting from the second term, each term is formed by multiplying the previous term by 2 and then taking the remainder when divided by 10. What is the 2022nd term?

（A）1　　（B）2　　（C）4　　（D）6　　（E）8

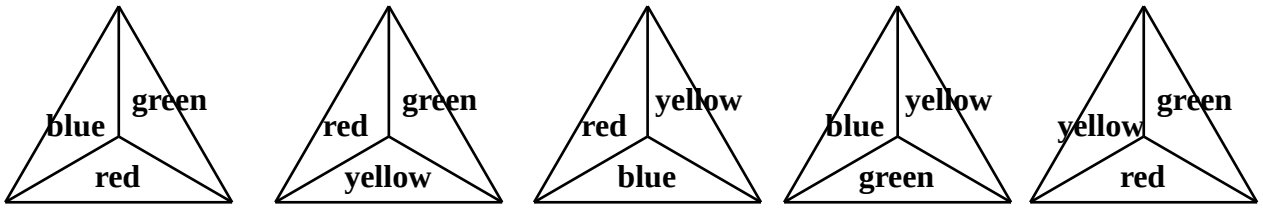
Answer：_____

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5. A tetrahedron is a regular pyramid made up of four equilateral triangles. It is possible to paint each face with a different colour --- red, green, yellow and blue. Five different views of the tetrahedron are shown below. Which among the choices is inconsistent with the other four?



（A）　（B）　（C）　（D）　（E）



Answer：_____

Questions 6 to 13, 5 marks each

6. One day, Robert went to a library 1600 meters away from home to meet with his classmates at 13 : 20. Robert leaves home at 13 : 00, walking at a constant speed of 80 meters per minute. At exactly 13 : 05, Robert's mother noticed that he left his wallet at home, so she chased him walking at a speed of 120 meters per minute. Robert only realized he left his wallet at home at 13 : 10, so he walked back casually at a speed of 80 meters per minute. Robert and his mother met halfway through. After taking his wallet, he again walked back towards the library at a speed of 80 meters per minute. If we ignore the time Robert took to get the wallet from his mother, how much time, in minutes, did he waste?

Answer : _____ minutes

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7. Mr. Andrew's office is located in City *T*, while he lives in City *S*. Every day after timing-off from work, he takes the same train to return to his home city. The train station in city *S* is still some distance away from his home, and his wife travels by car to the train station to pick him up. It is known that the train and his wife arrive at the train station at the same time. One day, due to some reasons, his wife left their house 30 minutes later than usual. Upon arriving at the train station in City *S*, Andrew noticed that his wife hasn't arrived yet so he decided to walk home. On his way walking home, he met his wife along the way and rode the car immediately. Upon reaching home, Andrew observed that they arrived home 22 minutes later than usual. How long, in minutes, did Andrew walk?



Answer : _____ minutes

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8. There are four cards with digits 0, 1, 2 and 3 written on each of them. Choose three different cards and arrange them to form all possible three-digit numbers, where 0 is not allowed be leading digit. What is the sum of all those three-digit numbers that are odd?

Answer : _____

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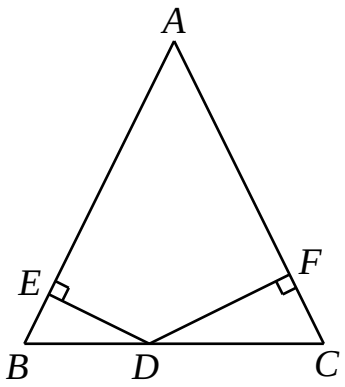
9. Three positive 9-digit numbers *A*, *B* and *C* each contains all the digits from 1 to 9 exactly once. If $A - B = C$, then what is the largest possible value of integer *C*?

Answer : _____

10. In a line, there is at least one girl and at least one boy. Two children with exactly 4 or 7 other children between them are of the same gender. What is the maximum number of children in this line?

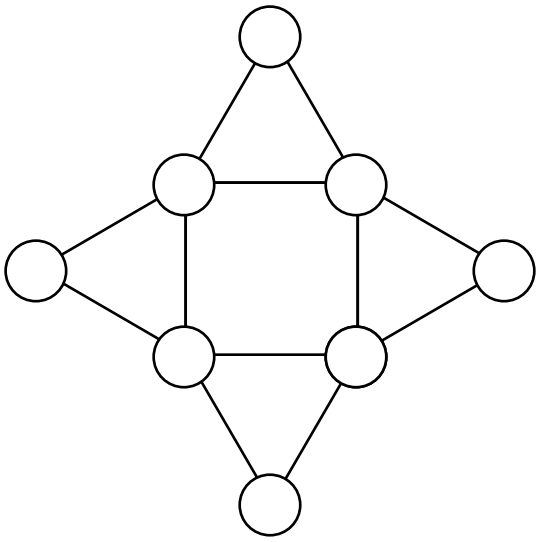
Answer : _____ children

11. In the diagram below, ABC is an isosceles triangle with $AB = AC$ and $BC = 48$ cm. If $DE : DF = 5 : 7$, what is the length, in cm, of DC?



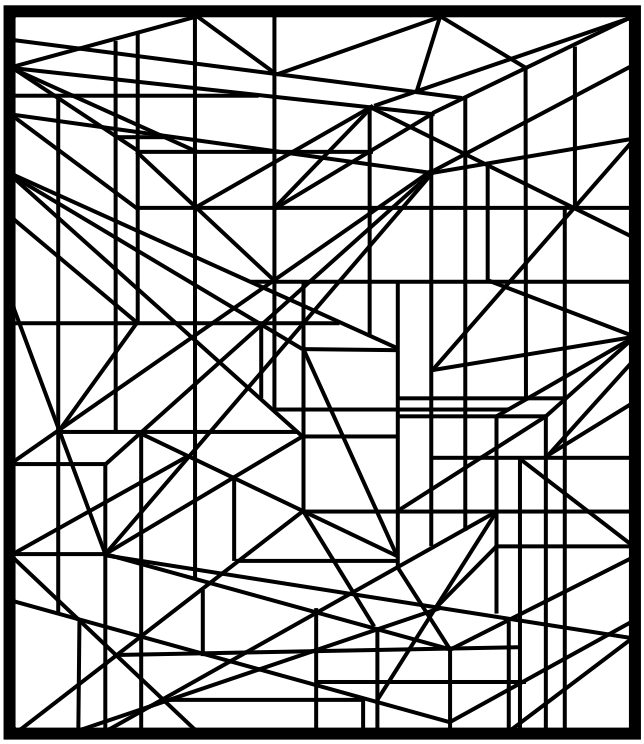
Answer : _____ cm

12. Numbers 1, 2, 3 and 4 are to be placed into the circles in the diagram below, where each circle must be filled with exactly one number. If two circles connected by a line should not contain the same number, find the total number of ways in filling the diagram.



Answer : _____ ways

13. How many cubes which are shown in Figures 1 and 2 are found inside the rectangular pattern below?



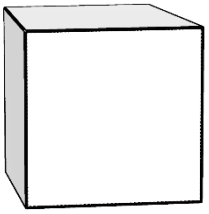


Figure 1

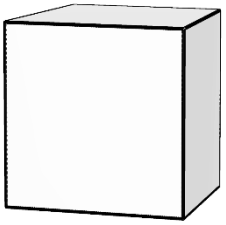


Figure 2

Answer： _____ cubes

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. A continuous look-and-say sequence is formed this way: Start with three 1's; read the terms as "three 1's" and write down what you read as "31" at the end of the sequence; read the next group of unread identical numbers as "one 3" and write "13" at the end; continue this process to get an infinite sequence 1, 1, 1, 3, 1, 1, 3, 2, 1, ⋯.
    (a) What is the largest possible digit you could write down? Why? (5 points)
    (b) What is the 20th term of this sequence? Why? (5 points)
    (c) Will the number "333" appear in the sequence? Why or why not? (10 points)

Answer： (a) The largest possible digit is _____ . The reason is:

(b) The 20th term of this sequence is _____ . The reason is:

(c) ☐ Yes. ☐ No. The reason is:

15. Divide the numbers 1, 2, …, 15 into five groups, such that each group contains three numbers and sum of numbers in each group is 24. Please provide an example.

Answer：

Group 1: { , , }	Group 2: { , , }
Group 3: { , , }	Group 4: { , , }
Group 5: { , , }	